# TA Tutorial Session — The ML Workflow & Habits

| **Faculty Session Reference** | **Academic Session - 13 - The ML Workflow & Habits, Assignment: The ML Workflow & Habits - Subjective, Assignment: The ML Workflow & Habits - Objective** |
|---|---|
| Estimated Duration | 90 minutes |

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## Opening Block (5 minutes)

> ***TA Instruction:*** *Run this block first. Set the tone, agenda, and connect to the main session. Keep it under 5 minutes.*

### Agenda Script

Use this or adapt it in your own words:

*"Welcome everyone. In today's tutorial session, we'll cover:

1. A quick topic summary to refresh what was covered in the faculty session.
2. 5 objective questions to test your understanding — vote in the chat.
3. 1 scenario-based practice question — these are directly aligned to your assessments, so pay close attention.
4. An open Q&A block where you can ask anything from the main session or today. Let's get started."*

### Context Setting

- **Link to main session:** In the faculty session, we explored the difference between rule-based AI and machine learning, and we discussed how to frame real-world datasets into features and labels for supervised learning models.
- **Why this matters for assessments:** Understanding how to correctly frame an ML problem, perform a train-test split, and evaluate a baseline model are core skills required to successfully complete your Colab notebook assignments.

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## Section 1: Topic-wise Summary (15 minutes)

> ***TA Instruction:*** *Cover all major topics briefly. Avoid deep explanations — focus on definitions, key logic, and use-cases. Ask 1–2 quick check questions to keep learners engaged. Target: 15 minutes total.*

### Why Are We Learning This? (Hook — Read Aloud Before Starting)

> *Imagine your phone automatically dims its screen when the battery drops below 15%. This creates a rigid response to a predefined condition. Rule-based AI relies on explicit instructions to handle simple, predictable tasks like this.*

> *Your manager asks you to forecast next year's sales based on thousands of past transactions. You cannot write simple if-else rules to predict complex numerical outcomes. Machine Learning and Regression models are designed to learn these patterns automatically from historical data.*

> *You are preparing for an important final exam by studying a textbook. You need to know if you are actually learning the concepts or just memorising the answers. The train-test split evaluates model learning just like taking a mock question paper before the real test.*

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### Topic 1: Rule-Based AI vs. Machine Learning

- Rule-based AI relies on human experts explicitly hardcoding rules (like if-else statements in Python) to tell the machine what to do and how to do it.
- Machine Learning gives computers the ability to learn without being explicitly programmed; the machine figures out the rules based on many input-output examples.
- Everyday examples of rule-based AI include washing machine wash modes, air conditioners, and fire alarms.
- Unstructured data (like images of mountains or elephants) cannot be resolved with static human rules, which is why Machine Learning is required to identify patterns.
- Both Rule-Based AI and Machine Learning fall under the broader umbrella of Artificial Intelligence (AI).

### Topic 2: Supervised Learning (Classification vs. Regression)

- Supervised learning is a type of Machine Learning where the outputs or labels are explicitly defined during training.
- The features (X) are the input variables, and the labels (Y) are the target outputs the model tries to predict.
- If the output trying to be predicted is a continuous numerical value (e.g., predicting next year's sales or house prices), it is framed as a Regression problem.
- If the output trying to be predicted is a categorical value (e.g., predicting if a credit card customer will default or not), it is framed as a Classification problem.

### Topic 3: The Machine Learning Lifecycle & Train-Test Split

- The dataset is divided into training data (typically 80%) to train the model, and testing data (typically 20%) to validate the model on unseen data.
- The training data acts as a textbook to learn the patterns, while the testing data acts as a mock question paper to evaluate that learning.
- Overfitting occurs when the training accuracy is very high, but the testing accuracy is very low, meaning the model failed to generalize.
- The `random_state` parameter ensures that the split between training and testing data remains constant and reproducible every time the code is run.

### Topic 4: Baseline Models and Evaluation

- A baseline model is a simple, dummy model used to benchmark the performance of actual machine learning algorithms.
- For a classification problem, predicting the most frequent (majority) class provides the baseline accuracy floor.
- For a regression problem, predicting the average value serves as the baseline performance.
- Real machine learning is an iterative process where you try different algorithms (like logistic regression or gradient boosting) and select the one that beats the baseline test accuracy.

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## Section 2: Objective Questions (15 minutes)

> *▯* ***TA Instruction:*** *Share screen with one question at a time. Ask learners to answer in chat or vote. After each answer, explain in 30–60 seconds — why correct is correct, why each wrong option is wrong. Keep explanations tight.*

### Question 1

Neha's Python script turns on the AC at 26C and lights after 7 PM. Which AI paradigm does her script best represent?

- A) Supervised Machine Learning
- B) Unsupervised Learning
- C) Rule-based AI
- D) Reinforcement Learning

**Correct Answer: C**

- A — Wrong. Supervised ML learns patterns from given data; this script uses pre-written explicit logic.
- B — Wrong. Unsupervised learning finds hidden patterns without explicit labels; this script has explicit rules.
- C — Correct. The script consists entirely of explicit if-else conditions written by a human, making it Rule-based AI.
- D — Wrong. Reinforcement learning is an external concept not applicable here; the script doesn't learn from experience.

> *Source: Assignment: The ML Workflow & Habits - Objective*

### Question 2

A teacher wants to predict if a student will Pass or Fail based on attendance, study hours, and mock score. Which column is the label (y)?

- A) attendance
- B) study_hours
- C) mock_score
- D) result

**Correct Answer: D**

- A — Wrong. Attendance is an input feature (X) used to find a pattern.
- B — Wrong. Study hours are an input clue used to make the prediction.
- C — Wrong. Mock score is a feature (X) used by the model.
- D — Correct. The label (y) is the target column the model is trying to predict, which is the Pass/Fail result.

> *Source: Assignment: The ML Workflow & Habits - Objective*

### Question 3

Priya has 5,000 rows. She does a test_size=0.20 split, then splits the rest with test_size=0.25. How many rows will the validation set have?

- A) 500
- B) 750
- C) 1,000
- D) 1,250

**Correct Answer: C**

- A — Wrong. 500 would only be 10% of the original dataset.
- B — Wrong. 750 does not represent the correct mathematical split.
- C — Correct. The first split leaves 4,000 rows. 25% of those 4,000 rows equals exactly 1,000 validation rows.
- D — Wrong. 1,250 would be 25% of the original 5,000, but the second split happens on the remaining 4,000.

> *Source: Assignment: The ML Workflow & Habits - Objective*

### Question 4

Rohit's DummyClassifier gets 68% accuracy on a test set where 68% of labels are "Pass". What best explains this result?

- A) The model learned patterns from 68% of training data
- B) The model correctly identifies difficult test edge cases
- C) It always predicts "Pass" and reflects class distribution
- D) The model was evaluated on the 68% of training data

**Correct Answer: C**

- A — Wrong. A DummyClassifier does not learn patterns from the training features.
- B — Wrong. The baseline dummy model does not evaluate complex edge cases.
- C — Correct. The dummy model simply predicts the majority class ("Pass") every single time without learning anything.
- D — Wrong. The model was evaluated on the test set, but it achieved 68% simply because the test set contains 68% "Pass" labels.

> *Source: Assignment: The ML Workflow & Habits - Objective*

### Question 5

Zara runs Python ML code on her basic budget laptop using Google Colab and the output appears instantly. Why is it so fast?

- A) Her laptop has a hidden Python turbo-boost feature
- B) Colab executes code on Google's cloud servers, not her laptop
- C) Colab compresses code before running it on any machine
- D) Python automatically optimises itself on budget laptops

**Correct Answer: B**

- A — Wrong. Laptops do not possess a hidden Python turbo-boost feature.
- B — Correct. Google Colab provides a remote runtime, meaning the heavy computation executes on Google's cloud servers.
- C — Wrong. Code compression is not the mechanism that makes Google Colab fast.
- D — Wrong. Python does not automatically optimize hardware performance on budget laptops.

> *Source: Assignment: The ML Workflow & Habits - Objective*

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## Section 3: Scenario-based Practice Questions (40 minutes)

> 💡 ***TA Instruction:*** *This is the core section. For each question: (1) Read the scenario aloud, (2) Give learners 5–8 minutes to attempt, (3) Ask "How would you start?" before revealing the answer, (4) Walk through the solution step by step. Do NOT jump to the answer immediately.*

### Question 3.1: StreamNow Subscription Churn Baseline *(~40 mins)*

**Scenario:** StreamNow, a music streaming platform, wants to know: "Based on a user's listening behaviour this month, will they cancel their subscription next month?" You are given a dataset representing user behaviour containing columns like `avg_daily_listening_mins` , `unique_artists_played` , `days_app_opened` , and a unique `user_id` . The outcome is recorded in a column called `will_cancel` (Yes / No).

**Task:** Frame the ML problem, identify the features (X) and label (y), write the code logic to perform a 60/20/20 train/validation/test split, and establish the performance floor using a DummyClassifier. Finally, interpret what a baseline accuracy means for this dataset.

**Answer:**

**Step-by-step approach:**

1. Identify the input features (X) and the target label (Y). Remember to drop identifiers that hold no predictive value.
2. Determine if this is a Classification or Regression problem based on the label type.
3. Use `train_test_split` sequentially to create a 60% training, 20% validation, and 20% test split. Fix the random state.
4. Fit a `DummyClassifier` tracking the most frequent class to establish your baseline accuracy.

**Model Answer:**

**Problem Framing:** Given `avg_daily_listening_mins` , `unique_artists_played` , and `days_app_opened` , I want to predict `will_cancel` (Yes / No). This is a **Classification** problem because the output is a discrete category, not a continuous number.

**Code Logic:**

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.dummy import DummyClassifier
from sklearn.metrics import accuracy_score, classification_report

# 1. Separate Features and Label
# Drop the unique identifier as it contains no predictive signal
X = df.drop(columns=["user_id", "will_cancel"])
y = df["will_cancel"]

# 2. Train / Validation / Test Split (60 / 20 / 20)
# First split: Carve out 20% for the test set
X_temp, X_test, y_temp, y_test = train_test_split(X, y, test_size=0.20, random_state=42)

# Second split: Split remaining 80% into 75% train (60% total) and 25% val (20% total)
X_train, X_val, y_train, y_val = train_test_split(X_temp, y_temp, test_size=0.25, random_state=42)

# 3. Build and Evaluate the Baseline
baseline = DummyClassifier(strategy="most_frequent", random_state=42)
baseline.fit(X_train, y_train)
```

```
# Predict on test set
baseline_preds = baseline.predict(X_test)

# Measure baseline accuracy
baseline_acc = accuracy_score(y_test, baseline_preds)
print(f"Baseline Accuracy: {baseline_acc:.2%}")
print(classification_report(y_test, baseline_preds))
```

**Interpretation:** If the baseline accuracy is around 70-75%, this simply reflects the proportion of "No" (non-cancellation) cases in the dataset. The baseline model has learned nothing; it just predicts the majority class. If you check the classification report, the recall for the "Yes" class will be 0% because the baseline never correctly predicts a cancellation. Any real ML model built afterward must beat this baseline accuracy floor and achieve meaningful recall for the "Yes" class to be useful to the business.

> *Source: Assignment: The ML Workflow & Habits - Subjective*

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## Section 4: Q&A Block (20 minutes)

> *🎤 **TA Instruction:** Open the floor. Prioritise conceptual doubts and assessment-related queries first. If unsure of an answer, acknowledge and commit to following up — do not guess.*

### How to Run This Block

- Ask: *"Any questions from the faculty session or from today's questions?"*
- If no questions immediately: prompt with *"Which question felt hardest? Let's revisit it."*
- If a question is outside the session scope: *"That's a great question — it goes beyond today's session. I'll check and follow up. Let's note it."*

### Anticipated Doubt Areas

- The exact difference between features (X) and labels (Y) and why we drop identifiers like `user_id`.
- The distinction between a Classification problem and a Regression problem based on business scenarios.
- Why we need both a training set and a testing set, and the risk of overfitting.
- What `random_state` actually does when running a train-test split in code.
- Why a high baseline accuracy (e.g., 98.5%) in a highly imbalanced dataset can be completely misleading.

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